www.csvtuonline.com

Printed Pages – 7

328812 (28)

BE (8th Semester)

Examination, April-May, 2013

Branch : Et & T

VLSI DESIGN

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) All questions carry equal marks.

(ii) Attempt all questions.

(iii) Part (a) of each question is compulsory.

(iv) Attempt any two parts from b, c and d of each question.

(v) Draw suitable diagram wherever required.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Q. 1. (a) Write the year in which VLSI era was started. 2

(b) With suitable example explain divide and conquer technique for VLSI Design. 7

(c) Assume that a gate array contains the type of logic cell shown in fig no. 1. The inputs IN_1, IN_2, IN_3 ..... IN_7 can be connected to either "1" or "0" or any logic signal. Show how the logic cell can be used to realize the function $f = x_1x_2 + x_2x_3$. 7

www.csvtuonline.com

www.csvtuonline.com

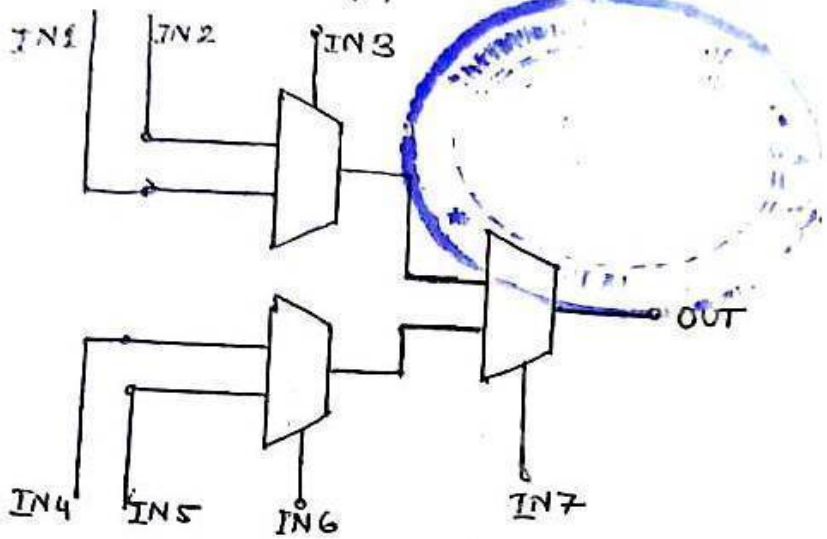

Fig. 1

(d) Realize the function

$f = x_1x_2x_4 + x_2x_3\bar{x}_4 + \bar{x}_1\bar{x}_2\bar{x}_3$ using 7, two input LUTs. Label the output of each LUT with an expression representing the logic function that it implements. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Q. 2. (a) Represent the buried contact for CMOS design rule. 2

(b) With suitable diagram explain the transistor design rules for CMOS. 7

(c) Draw the stick diagram for CMOS, 2 input NOR gate. 7

(d) Calculate the ON resistance of a circuit shown in fig no. 2 from V_{DD} to GND if n channel sheet resistance $R_{sn} = 10^4\ \Omega$ per square and p channel sheet resistance $R_{sp} = 2.5 \times 10^4\ \Omega$ per square. 7

www.csvtuonline.com

www.csvtuonline.com

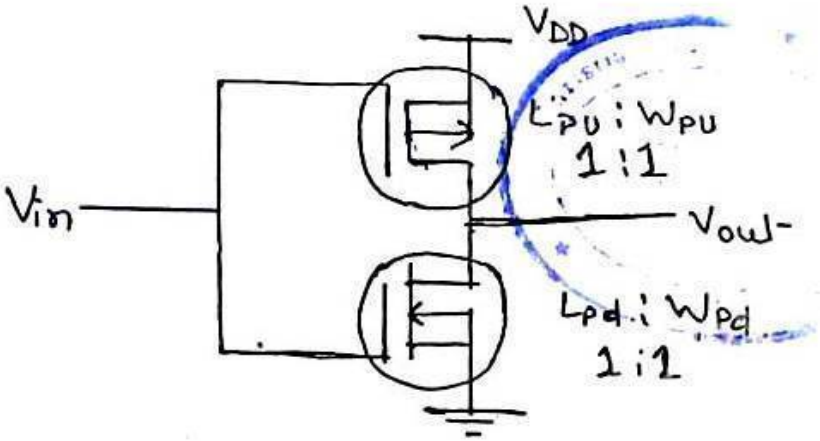

Fig. 2

Q. 3. (a) What is true single phase clocked register ? 2

(b) Draw the stick diagram of 6T RAM cell. 7

(c) Construct stick encoding for '1' bit adder. 7

(d) Draw the stick diagram of JK flip flop. 7

Q. 4. (a) What are different types of modeling in VHDL ? 2

www.csvtuonline.com

www.csvtuonline.com

(b) Write a VHDL code for the circuit shown in the fig no. 3. 7

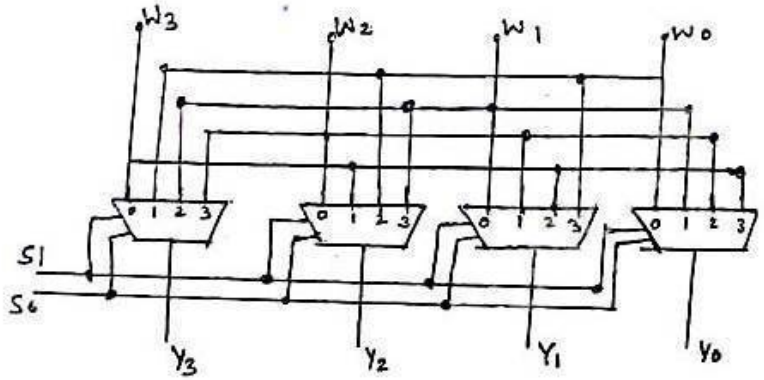

Fig. 3

(c) Design a system which counts one digit decimal numbers and displays the result to the seven segment display connected to it. 7

(d) Write a VHDL code to find the factorial of a number which is available in the binary format at the system input. 7

www.csvtuonline.com

www.csvtuonline.com

Q. 5. (a) What is Moore FSM ? 2

(b) Explain how BIST is useful for testing and verification for half adder. 7

(c) Design a test bench for parallel in serial out shift register. 7

(d) Write a structural code in VHDL of a sequence detector 1101, synthesized by using mealy machine. 7

www.csvtuonline.com

www.csvtuonline.com